ModelIt!

Student Activity Pack

Superbugs: Evolution You Can See

Why Antibiotics Are Losing the War Against Bacteria

Grade 8 | NGSS MS-LS4-4, MS-LS4-6 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

Multiple Choice: Circle the best answer for each question.

1 What is the primary function of antibiotics?

- **A.** To kill or stop the growth of bacteria
- **B.** To boost the human immune system
- **C.** To kill viruses that cause the common cold
- **D.** To prevent all types of infections permanently

2 Where do new traits in a population originally come from?

- **A.** Organisms choose to develop traits they need to survive
- **B.** Random changes (mutations) in an organism's DNA
- **C.** The environment directly changes the organism's genes
- **D.** Traits are created by natural selection itself

3 In the process of natural selection, which organisms are most likely to survive and reproduce?

- **A.** The largest and strongest organisms
- **B.** Organisms with traits that are best suited to their current environment
- **C.** The oldest organisms in the population
- **D.** Organisms that can change their traits when the environment changes

4 A doctor prescribes antibiotics for a patient with a bacterial infection. The patient feels better after 3 days but has a 10-day prescription. What should the patient do?

- **A.** Stop taking the antibiotics since the infection is gone
- **B.** Take the remaining pills all at once to finish faster
- **C.** Complete the full 10-day course as prescribed
- **D.** Save the remaining pills for the next time they get sick

1 What are antibiotics and what do they do?

2 Why do you think some infections are harder to treat than others?

3 What does 'survival of the fittest' mean to you?

4 **Draw what you think happens when bacteria meet an antibiotic.**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Antibiotic Use | Mutation Rate | Resistant Bacteria | Treatment Effectiveness

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Treatment

Moderate antibiotic use, normal mutation rate

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Antibiotic Overuse

Lock Antibiotic Use to 100%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Responsible Use

Set Antibiotic Use to 30%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Horizontal Gene Transfer — Bacteria can share resistance genes with nearby bacteria

Immune System Strength — A stronger immune system can fight bacteria alongside antibiotics

Biofilm Formation — Bacteria form protective communities that resist treatment

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are bacteria becoming impossible to kill? Use evidence from your simulation to support your answers.

1. **Why is it dangerous to stop taking antibiotics when you feel better but before the course is done?**

2. **How is antibiotic resistance an example of natural selection happening in real time?**

3. **Why might using antibiotics in livestock farming increase resistance in human diseases?**

4. **What would happen to modern medicine if antibiotics stop working completely?**

5. **How does your model show that evolution is NOT random, even though mutations ARE random?**

8 STEM Challenge: Design a Hospital Antibiotic Policy

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YOUR ENGINEERING MISSION

Design an evidence-based antibiotic use policy for a hospital that minimizes resistance evolution while still effectively treating infections.

THE SCENARIO

A hospital is seeing increasing antibiotic-resistant infections in patients. The medical director needs your team to propose a new policy using evidence from your model.

Engineering Design Thinking

1. WHAT GUIDELINES WILL YOU SET FOR WHEN ANTIBIOTICS SHOULD BE PRESCRIBED?
2. HOW WILL YOU MONITOR RESISTANCE LEVELS IN THE HOSPITAL?
3. WHAT EDUCATION PROGRAM WILL YOU CREATE FOR PATIENTS?
4. HOW WILL YOU MEASURE IF YOUR POLICY IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Epidemiologists and Microbiologists track antibiotic resistance patterns worldwide and develop strategies to combat superbugs. They earn $80,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In a population of bacteria exposed to antibiotics, resistant bacteria survive while non-resistant bacteria die. Which statement best explains why the resistant bacteria survived?

- **A.** The antibiotics caused the bacteria to mutate into resistant forms
- **B.** The resistant bacteria chose to activate their defense genes
- **C.** Random mutations that occurred before antibiotic exposure happened to provide resistance
- **D.** The resistant bacteria learned to fight the antibiotic through repeated exposure

2 A model shows that when antibiotic use is set to 100%, resistant bacteria reach 90% of the population in 10 generations. When antibiotic use is set to 30%, resistant bacteria reach only 25% in the same time. What does this evidence support?

- **A.** Lower antibiotic use prevents all mutations from occurring
- **B.** Higher antibiotic use creates a stronger selective pressure, accelerating resistance evolution
- **C.** Resistant bacteria only appear when antibiotic use is above 50%
- **D.** Antibiotics at 30% are too weak to kill any bacteria

3 A hospital notices that MRSA infections have increased by 300% over 10 years. Using your understanding of natural selection, which factor most directly drove this increase?

- **A.** Bacteria deliberately evolved to resist methicillin
- **B.** Repeated antibiotic use selected for bacteria carrying pre-existing resistance genes over many generations
- **C.** The hospital became dirtier over time, creating new resistant bacteria
- **D.** Patients' immune systems became weaker, allowing bacteria to mutate faster

4 Your model includes Mutation Rate and Antibiotic Use as external components and Resistant Bacteria and Treatment Effectiveness as internal components. If you wanted to slow the rise of resistant bacteria, which strategy would your model support?

- **A.** Increase mutation rate to create bacteria that are too mutated to survive
- **B.** Use antibiotics at maximum levels to kill all bacteria before they can reproduce
- **C.** Reduce antibiotic use to decrease the selective pressure favoring resistant bacteria
- **D.** Eliminate all bacteria from hospitals using stronger chemicals